# Sensors

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***Abstract*— This report explores sensors. For each component it explains the working principle, data processing typical errors and fixes, integration in the car, and indicative cost and sourcing with a preference for Indian suppliers. The components covered are a Hesai Pandar40P LiDAR, a Stereolabs ZED 2i stereo camera, a VectorNav VN-200 GNSS/INS, wheel speed sensors, an NVIDIA Jetson AGX Orin computer, and steer-by-wire and brake-by-wire , accelerometer, magnetometer, Hall-effect and optical sensing technologies used inside them.**

***Keywords— LiDAR, stereo camera, GNSS/INS, IMU, wheel speed sensor, Hall effect, Jetson AGX Orin, steer-by-wire, brake-by-wire***

## INTRODUCTION

LiDAR and cameras perceive the track, a GNSS-aided inertial navigation system and wheel speed sensors estimate the car's own motion, a computer runs the software, and steering and brake actuators carry out the commands. This report describes the sensors and actuators work, how its data is processed, which errors to expect and how to handle them, and what it costs and where it can be sourced.

The components covered are a Hesai Pandar40P LiDAR, a Stereolabs ZED 2i stereo camera, a VectorNav VN-200 GNSS/INS, wheel speed sensors, an NVIDIA Jetson AGX Orin computer, and steer-by-wire and brake-by-wire actuation. The underlying accelerometer, magnetometer, Hall-effect and optical sensing technologies are also explained, because they appear inside several of these devices.

## LIDAR

LiDAR (Light Detection and Ranging) measures distance by emitting a short laser pulse and timing how long the reflection takes to return to the receiver. Because light travels about 30 cm per nanosecond, the range follows from the time of flight Δt:

$$d = (c \times \Delta t) / 2$$

where c is the speed of light and the factor of two removes the return trip. A sensor therefore needs sub-nanosecond timing to reach centimeter accuracy. Each laser firing also has a known azimuth θ and elevation φ, so the measured range R converts to a point in the sensor frame:

$$x = R \cos\varphi \cos\theta, \quad y = R \cos\varphi \sin\theta, \quad z = R \sin\varphi$$

One full rotation produces a frame of thousands of such points, called a point cloud, and each point also carries a single reflectivity value and a timestamp. Automotive LiDAR uses invisible near-infrared light, commonly 905 nm with inexpensive silicon detectors or 1550 nm, which permits higher eye-safe power and longer range at higher cost. Scanning can be mechanical (a spinning stack of emitters), MEMS-mirror based, or solid-state. A 2D LiDAR scans a plane and suits small indoor robots, while a 3D LiDAR stacks several beams or scans a volume and is needed for real driving perception.

LiDAR gives direct, accurate geometry, works at night because it carries its own illumination, and does not depend on texture. Its weaknesses are sparse data at long range, degraded returns in heavy rain, fog and dust, confusion on mirrors and glass, weak returns from very dark surfaces, and the absence of color. The last point matters for us, because cone color marks the left and right track boundaries and a LiDAR cannot see it.

LiDAR data goes through the same chain of steps for every frame, roughly every 50 to 100 ms.

First, the sensor's packets are converted into 3D points. The sensor takes about 0.1 s to spin once, so at 20 m/s the car moves 2 m during a single scan. To correct this, each point is shifted using the car's speed and turn rate, as if the whole scan had been taken at one instant. This is called deskewing.

Next, the cloud is cleaned. It is cropped to the area that matters, thinned by replacing the points inside each small cube (10 to 20 cm) with their average, and filtered to remove isolated noise points. The car's own body is cut out as well.

The road surface is then removed. Otherwise, objects standing on it would be joined together through the ground points. On slopes, methods that fit several local planes are used.

Finally, the remaining points are grouped into objects. Points within a set distance of each other go into the same cluster, and clusters that are too small or too large are discarded. If the distance is too small, one cone splits into two clusters. If it is too large, neighbouring cones merge, so the distance is usually increased with range. Each cluster gives a position and size, and for cones, the size, point count and reflectivity decide whether it really is a cone. The camera then supplies the colour, and the cones are added to a map using the car's position.

## HESAI PANDAR 40P

The Pandar40P is a 40-channel mechanical LiDAR. Its manual describes multiple emitter and receiver pairs mounted on a motor that rotates through 360°.

Its cost generally exceeds ₹3,00,000, and it is usually procured through authorized distributors or industrial import channels.

## ZED 2I CAMERA

A mono camera has one lens and produces a flat image. It can identify what something is, but not how far away it is, because every point on a viewing ray lands on the same pixel. A stereo camera has two lenses a known distance B apart. The same object appears at slightly different horizontal positions in the left and right images, and this shift, called the disparity d, gives the depth:

$$Z = f \cdot B / d$$

where f is the focal length in pixels. A near object has a large disparity and a far object a small one. Because the disparity is measured to a finite precision, the depth error grows roughly with the square of the distance, so stereo is accurate close to the car and increasingly uncertain far away.

Camera processing follows several steps. Images are first undistorted using the intrinsic calibration, which consists of the focal lengths, the image center and the lens distortion coefficients, found by photographing a checkerboard from many angles. A neural network detector such as YOLO then returns boxes and class labels. Each detection keeps the timestamp of its image to ensure accuracy.

the camera supplies cone colour, a second independent distance estimate close to the car, and redundancy if the LiDAR misses a thin or distant object.

The Stereolabs ZED 2i is a stereo camera used for depth estimation, object detection and visual perception in autonomous vehicles. It can be purchased through Indian suppliers such as MG Super Labs and Fab.to.Lab. The estimated cost is approximately ₹73,000, depending on availability and supplier pricing.

## IMU AND INS

An inertial measurement unit (IMU) is only the measuring hardware: gyroscopes for rotation rate, accelerometers for acceleration, and sometimes magnetometers. It outputs raw readings and does not know where it is or which way it points.

An inertial navigation system (INS) adds further computation that integrates the readings, removes gravity, and outputs orientation, velocity and position. When a GNSS receiver is added to correct drift, the result is a GNSS-aided INS, which is the category of the VN-200.

## VECTORNAV 200

The VN-200 is a GNSS-aided INS. The two sensor groups complement each other. An IMU is fast and smooth, but it drifts because it works by integrating, so small errors accumulate. GNSS does not accumulate error, but it is slow, noisy and unavailable in tunnels. The Kalman filter predicts the motion from the inertial data and corrects it whenever a GNSS measurement arrives, which gives a smooth estimate that does not drift.

VN-200 has four main uses. It provides the car's own motion so that LiDAR scans can be deskewed. It gives LiDAR scan matching an initial pose guess, which matters because matching can lock onto a wrong solution if it starts too far from the truth. It lets the tracker subtract the car's own velocity, so that stationary cones are recognized as stationary. And its pitch and roll can be used to compensate for the car's tilt during ground removal.

VectorNav's default axes (x forward, y right, z down) differ from the usual ROS convention (x forward, y left, z up), so a conversion is needed

When imported through Indian engineering distributors, the price typically ranges between ₹3,000,000 to ₹4,50,000 INR per unit for standard requirements. This variance includes international freight, customs duties, and local dealer margins.

## NVIDIA JETSON AGX ORIN

It's the computer that runs everything (ROS2, Neural Networks, etc.) It receives data through physical **ports**. Each sensor plugs into the port that matches how it sends data.

.Hard real-time work such as counting wheel pulses and running the fast steering loop is not placed on the Jetson, because Linux cannot guarantee exact timing. It is delegated to a microcontroller or the actuator's own controller. The computer must also be kept cool, powered from a stable converter and mounted to withstand vibration.

The NVIDIA Jetson AGX Orin is an embedded computing platform used to process sensor data and execute artificial intelligence algorithms in autonomous vehicles. It supports tasks such as object detection, sensor fusion and path planning. According to the reference report, it can be sourced through distributors serving India, including Arrow Electronics and Mouser India, at an estimated cost of ₹4,00,000–₹5,50,000 per unit.

## WHEEL SPEED SENSORS

A wheel speed sensor counts pulses as a toothed ring or magnet ring turns with the wheel. More pulses per second means a faster wheel. Hall-effect sensors work down to very low speed. Pulses can be counted directly by a microcontroller with interrupts and a hardware timer, which is preferred because a busy Linux computer cannot guarantee the precise timing.

If the ring has N pulses per revolution and P pulses are counted in a window $\Delta t$, the wheel speed and the car speed for wheel radius r are:

$$\omega = 2\pi P / (N \Delta t), \quad v = \omega r$$

At low speed the window contains few pulses and the reading becomes coarse, so the time between pulses is measured instead.

The ZF (Cherry) GS100502 is an active Hall-effect sensor suitable for measuring wheel rotational speed. It is designed for automotive environments and can be sourced through distributors such as Element14 India or Mouser India. Its estimated cost is approximately ₹3,300 per sensor. A vehicle using one sensor for each wheel would require four units, giving an estimated total of ₹13,200.

## STEER-BY-WIRE AND BRAKE-BY-WIRE

*Steer-by-wire*

In a conventional car the driver's hands and feet are mechanically linked to the wheels and brakes. In a by-wire system an electronic signal replaces some of that link: a

controller sends a command and an electric actuator does the physical work. The data flow is as follows. The planner produces a path with target speeds. The localization provides the car's pose, and the speed comes from the wheel sensors and the VN-200. Inputs are first validated: stale timestamps, impossible position jumps, an empty path or high localization uncertainty all cause the controller to refuse to steer blindly.

*Brake-by-wire*

Brake-by-Wire is an electronically controlled braking system in which braking commands are transmitted through electrical signals rather than relying entirely on a direct mechanical or hydraulic connection between the brake pedal and the braking mechanism. In a driverless car, this technology allows the autonomous control software to control the braking system without requiring the driver to physically press the brake pedal. It plays an important role in regulating vehicle speed, slowing down before turns and bringing the vehicle to a stop when necessary.

When perception and planning show that the car must slow or stop, the controller uses the current speed and the available stopping distance to calculate the required deceleration.

## ACCELEROMETER

An accelerometer measures specific force, the acceleration it feels including gravity. A car at rest on level ground therefore reads about 9.8 m/s² upward.

MEMS capacitive: A tiny silicon mass on springs moves when accelerated and changes the capacitance between itself and fixed plates. It is small, cheap and low in power, and it measures steady acceleration, so it can also sense tilt. It is the most common type, found in phones, cars and most IMUs.

MEMS piezoresistive: The moving mass bends a small beam, and the bending changes its electrical resistance. It tolerates high shock, which makes it useful for crash sensing, but it is more sensitive to temperature and noisier at low acceleration.

Piezoelectric: A crystal produces a voltage when the moving mass squeezes it. It is excellent for high-frequency vibration and very rugged, but it cannot measure steady acceleration, so it cannot sense gravity or tilt. It is used for vibration and engine testing rather than navigation.

Thermal: A heated gas bubble inside the chip shifts when the sensor accelerates, and temperature sensors detect the shift. It has no moving mass, so it survives shock well, but it responds slowly and has limited bandwidth.

Resonant: A tiny beam vibrates at a set frequency, and acceleration changes that frequency. It is very stable and accurate, but more complex and costly, and is used in higher-grade navigation and aerospace equipment.

Force-balance: An electromagnet holds the proof mass in place, and the current needed to hold it is the measurement. It gives excellent accuracy and low drift, but it is large, power-hungry and expensive, so it is used in military navigation and geophysics.

## MAGNETOMETER

A magnetometer measures magnetic field, and in a vehicle its main job is to sense the direction of the Earth's field, which gives a compass heading.

Hall effect: A current flows through a thin plate, and a magnetic field produces a small voltage across it. These sensors are cheap and small, but they have low sensitivity and are noisy. They suit strong fields, such as a magnet on a wheel, better than the Earth's weak field.

Fluxgate: A magnetic core is driven in and out of saturation, and the changes reveal the field. Fluxgates are accurate and low in noise, and they are the standard for precision work. They are also larger, more power-hungry and more costly, so they are used in aircraft, satellites and surveying.

Search coil (induction): A changing magnetic field induces a voltage in a coil. It is excellent for fast-changing fields, but it cannot measure a steady field, so it cannot work as a compass.

Magneto resistive: Measures magnetic fields by detecting changes in the electrical resistance of a material when it is exposed to a magnetic field. The resistance of the sensing element varies with the strength and direction of the applied magnetic field.

## HALL EFFECT

The Hall effect is the appearance of a small voltage across a thin current-carrying plate when a magnetic field passes through it at a right angle. The voltage grows with field strength and changes sign when the field reverses, so the sensor reports both strength and direction. Linear Hall sensors output a voltage proportional to the field and are used to measure position or current, while digital Hall switches and latches change state when the field crosses a threshold and are used to detect a passing tooth or magnet. In the car, Hall sensors count wheel pulses, sense pedal, steering and gear positions, commutate brushless motors and measure current without contact. They are contactless, work at zero speed, and tolerate dirt, oil and vibration well, but they have limited sensitivity, drift with temperature, and are disturbed by stray magnetic fields.

## OPTICAL SENSORS

Optical sensors pair an emitter, usually an LED or laser diode, with a detector such as a photodiode, phototransistor or image sensor. Through-beam sensors detect an object that interrupts the light, reflective sensors measure light returned from a nearby surface, and time-of-flight devices, including the LiDAR in this report, measure the return time.

Cameras are image-sensor arrays. The optical encoder is the form most relevant to us: a slotted disc passes through a through-beam sensor, and an incremental encoder produces pulses, with a quadrature version giving direction, while an absolute encoder has a unique pattern for every angle and reports the exact position even after power-up, which suits steering angle measurement.

## COMMON ERRORS

LiDAR: Scans smeared by vehicle motion are corrected with motion compensation using the VN-200 and wheel odometry. Noise from rain, dust and reflections is removed with outlier filters and, where available, the sensor's own filters. A single-plane ground fit fails on slopes, so zone-based or ring-based ground removal is used, and the sensor's tilt is corrected using the IMU. Far cones return too few points to be trusted, so a minimum point count or a camera confirmation

is required. Points close to the car are lost to the blind zone, and the car's own body appears in the scan unless it is masked. An incorrect mount height or tilt shifts every ground ring, which is checked by comparing the measured sensor height with the fitted ground plane.

Camera: Stereo depth error grows with the square of distance, so beyond a limited range the LiDAR is trusted instead, and disagreement between the two is treated as a warning. A drifted LiDAR-camera calibration is detected by the projection check and repaired by recalibration.

VN-200: Near buildings and trees, GNSS signals can bounce off surfaces, which shifts the computed position by several meters, or can be blocked entirely, which prevents a position fix. In either case the VN-200's position drifts, since its inertial sensors can only bridge a short gap before errors accumulate. The car then relies on wheel odometry and LiDAR scan matching to track its motion.

Wheel speed. Wheel slip is detected by comparing wheel speed with the IMU and the GNSS or LiDAR speed, and the filter is told to trust the wheels less. A wrong tire radius or pulses-per-revolution value is corrected by a measured-distance calibration, repeated when tires, pressure or load change. Coarse low-speed readings are solved by timing the gap between pulses. A dead sensor is flagged when one wheel reads zero while the others turn.

## CONCLUSION

This report examined the sensors and actuators that make up the driverless subsystem and how each one works, how its data is processed, which errors it produces and how those errors can be handled. The Hesai Pandar40P gives accurate 3D geometry of the cones, and the ZED 2i adds their colour and an independent distance estimate close to the car. The VN-200 and the wheel speed sensors tell the car how it is moving and where it is, the Jetson AGX Orin runs the software that combines them, and steer-by-wire and brake-by-wire turn the resulting decisions into motion. The accelerometer, magnetometer, Hall-effect and optical sensing principles behind these devices explain both their strengths and their limits. The cost and sourcing is also covered.

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